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Paleomagnetism of the Lower Devonian sandstones from Holy Cross Mountains, Central Poland: Part 1

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Abstract

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Paleomagnetic investigations of the Emsian quartz sandstones from Holy Cross Mountains (Central Poland) have revealed the presence of a three-component natural remanent magnetization. The viscous component of the NRM is carried by magnetite grains and is directed close to the present-day local geomagnetic field. The remaining two components are both carried by hematite, presumably of two-stage origin. Because it was not possible to apply a fold test, the acquisition time of each component could only be inferred indirectly. The hematite remanence of the highest (650 °C) blocking temperature is thought to be of post-tectonic (Permian) origin, as its direction before bedding correction ($D = 199°$, $I = -38°$) is almost identical with the characteristic NRM obtained (Lewandowski, 1983) from neighbouring Permian deposits. The intermediate component ($D = 231°$, $I = 0$, with a blocking temperature of around 610 °C), after applying a bedding correction, gives a paleomagnetic pole ($\lambda = 24°$ N, $\phi = 142°$ E) near to European Middle–Upper Devonian poles. Therefore, it is thought that this magnetization probably reflects the ambient geomagnetic field not very long after sedimentation.

Introduction

During the past few years, intensive paleomagnetic studies of the Devonian rocks have been made both from the European continent (Abrahamsen, 1977; Duff, 1980; Bachtadse et al., 1983; Perroud and Bonhommet, 1984) and the British Isles (Storetvedt and Carmichael, 1979; Storetvedt and Torsvik, 1983; Torsvik et al., 1983; Torsvik, 1984, 1985). Although unequivocal geotectonic interpretation of the results obtained so far is not possible, nonetheless some general conclusions have been drawn concerning geodynamic processes during the assembly of Pangea (e.g. Van der Voo, 1983). This paper reports paleomagnetic results from the Holy Cross Mountains (Central Poland). The investigation is part of the research program on the Upper Palaeozoic rocks from Norway (including Spitsbergen) and Poland, carried out by paleomagnetic groups in Bergen and Sogndal (Norway) and Warsaw (Poland). The Spitsbergen results have already been published (Løvlie et al., 1984; Jeleńska and Lewandowski, in press).

Geological setting

The Holy Cross Mountains form the northwest part of the Małopolska Massif and are situated just beyond the southwest border of the East European Platform. Two distinct environmental provinces (Fig. 1) have been distinguished from facies analysis (Czarnocki, 1957):

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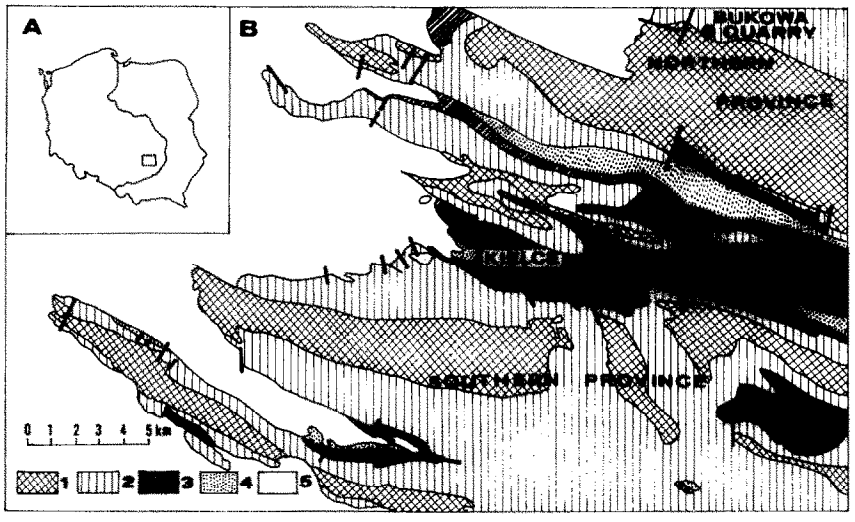


Fig. 1. Geological sketch map (B) of the western part of the Holy Cross Mts. (simplified after Szulczewski, 1971) with Bukowa Quarry localization (50.5° N, 20.3° E). Inset. A—Holy Cross Mts. on the background of Poland. *1*—Lower Palaeozoic, *2*—Lower–Middle Devonian, *3*—Upper Devonian, *4*—Lower Carboniferous, *5*—post-Variscan cover.

(1) deep water, in the Łysogóry region to the north (northern province);

(2) shallower water, in the Kielce region to the south (southern province).

In the south, because of continuous uplifting of the Kielce region during the Caledonian orogeny, a distinct disconformity and corresponding gaps are observed in the stratigraphic record, and Devonian deposition appears to have begun with Old Red Sandstone facies deposition in Emsian times. By contrast, sedimentation was continuous in the north, resulting in the deposition of both the shallow-water Downtonian facies and the overlying marine sediments of Emsian age.

For paleomagnetic purposes, four outcrops in the southern province (66 oriented cores and 7 hand samples) were sampled, together with a 100 m thick section of Emsian quartz sandstones in Bukowa Quarry in the northern province (33 hand samples). The Bukowa Quarry section consists of white and light grey quartz sandstones, interbedded with reddish silt–sandstones, particularly in the middle part of the sequence. The rocks show characteristic structures (ripple marks, fossil burrows and a rich brachiopod fauna) that suggest deposition in moderately shallow water, not far from the paleoshore line. The strata were tilted (25°–30° NNE) by Variscan tectonic movements, probably during Upper Carboniferous time (cf. Szulczewski, 1971).

In this paper only the results obtained from the Bukowa Quarry are presented, as too few samples from the southern province gave reliable results. It is expected that data from the southern province will be presented in the future, after re-sampling and further research.

Paleomagnetic measurements

Paleomagnetic measurements were made in the Paleomagnetic Laboratory of the Institute of Geophysics, Polish Academy of Sciences in Warsaw, using a Czech JR-4 spinner magnetometer. After being cut from the oriented hand samples (usually three specimens from the one sample), each cylindrical specimen was demagnetized by means of a thermal or AF (alternating field) cleaning method in order to remove secondary magnetization. However, most specimens (which numbered 96 in total) were exclusively thermally demagnetized, as this method proved to be more effective in removing secondary NRM components. The demagnetization procedure was carried out using a non-magnetic furnace in which the residual magnetic field was 10–15 gammas. Anisotropy measurements were made using a Czech KLY-2 kappa

bridge. The thermomagnetic method described by Kądziałko-Hofmokl et al. (1976), in which saturation remanence versus temperature is observed, was used to determine the nature of the magnetic carriers.

Results and discussion

Initial intensities of the NRM vectors were generally low and seldom exceeded 10^{-3} A m^{-1}. Nevertheless, all but 5 out of a total of 33 hand samples gave measurable paleomagnetic directions. During thermal cleaning, NRM intensities fell to 3×10^{-6} A m^{-1}, close to the apparatus noise level. Comparing the course of demagnetization paths, three groups of specimens can be distinguished, as follows:

(1) a normal polarity group, with NRM directions which remain in the Ist and IVth quadrants of the stereographic projection throughout the whole cleaning experiment;

(2) a reversed polarity group, occupying the 3rd quadrant of a stereonet, and

(3) a group of variable directions, the polarity of which changed from normal (initially) to reversed.

Examples of demagnetization behaviour typical of each group are presented in Figs. 2, 3 and 4. Kirchvink's (1980) method, in which straight segments on orthogonal plots are fitted to straight lines, was used to determine the NRM components. From a total of 96 specimens, the characteristic NRM components were determined in 77 cores. The complete results are presented in Fig. 5.

The magnetization directions obtained lie both in normal (N) and reversed (R) polarity sectors. The plots in Fig. 6 show the presence of both groups more distinctly. They enable each direction to be compared with its corresponding blocking temperature (T_b), measured as the maximum blocking temperature at which individual NRM components were removed. Figure 6 shows that the directions of R and N groups are recorded in grains of high and low blocking temperatures, respectively. Whereas the directions of group N show a Fisherian distribution, the R group is distributed in a non-Fisherian way, being drawn out along the equator. This spatial distribution of the R group directions could reflect several causes, for example:

(1) a rapid polar wander shift during depositon;

(2) hydrodynamic conditions during the deposi-

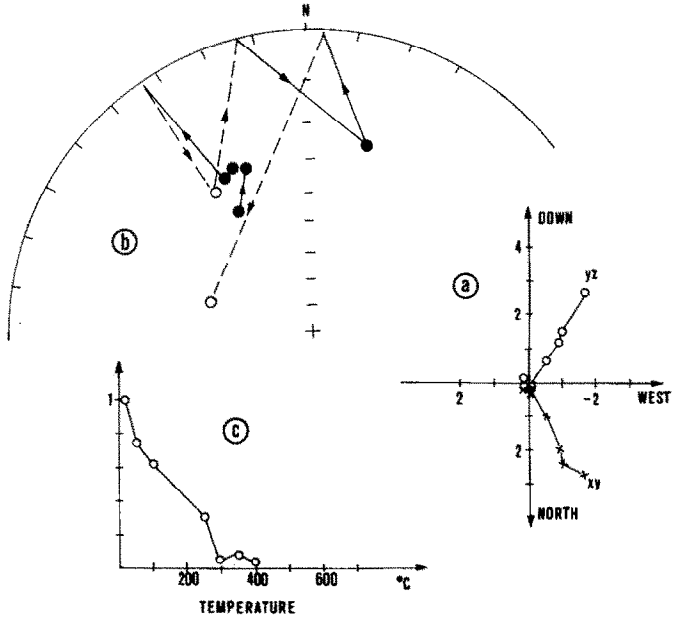


Fig. 2. (a) Orthogonal projection (magnetization units 8×10^{-5} A m^{-1}), (b) stereographic projection and (c) intensity plot of NRM behaviour during thermal demagnetization typical of group N—normal magnetization. Open (solid) stereonet symbols represent upper (lower) hemisphere directions. (After bedding correction.)

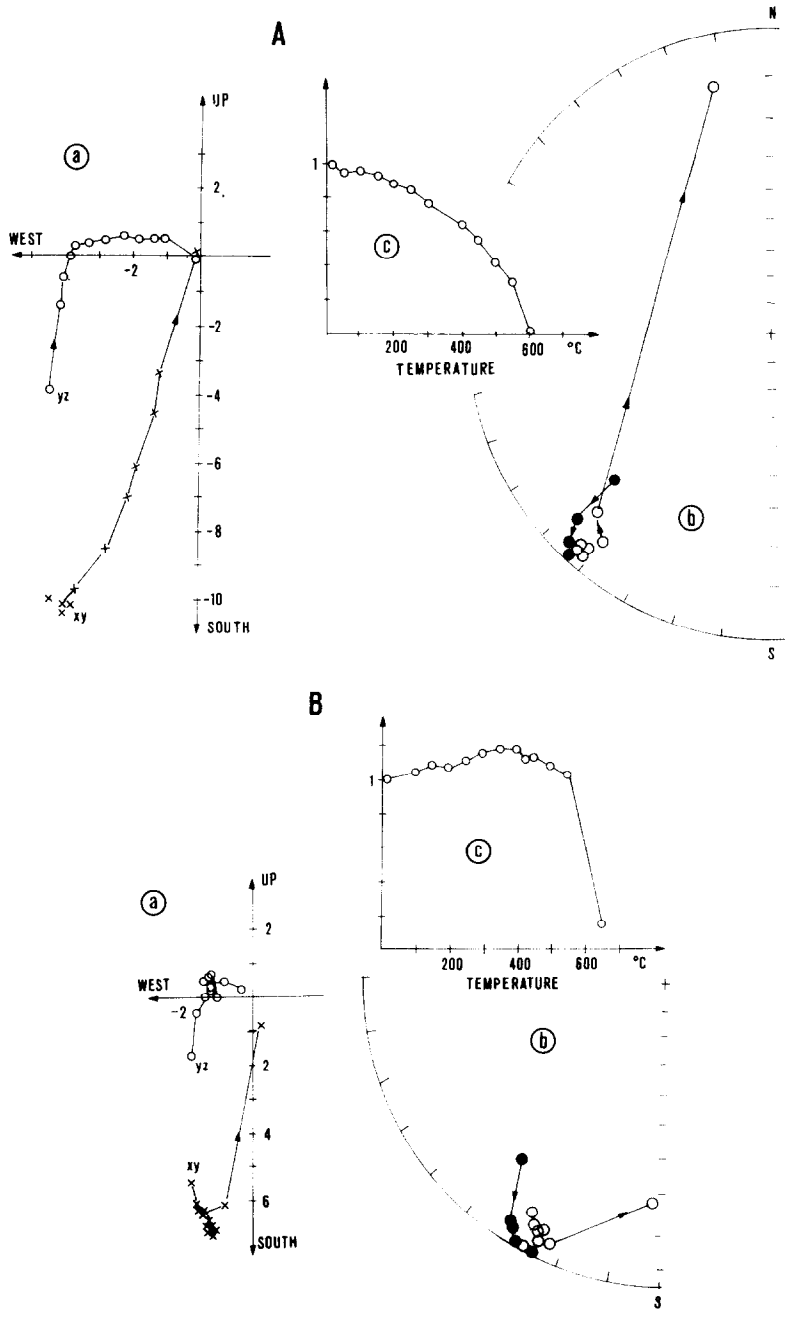


Fig. 3. NRM during thermal demagnetization of representative specimens of group R—reversed magnetizations. (A) RM subgroup; (B) RH subgroup. For explanation see Fig. 2. (After bedding correction.)

tion of sediment containing anisotropic magnetic grains (cf. Hamilton and Rees, 1970);

(3) inaccuracies of tectonic field measurements.

This last possibility is rejected, because the N group shows no such distribution. Any influence of underwater currents on anisotropic magnetic grains is also rejected, as the results of anisotropy investigations revealed random distribution of anisotropy axes. Polar wander shift during deposition should have resulted in systematic changes of characteristic NRM along the stratigraphic sequence, which is not the case (Fig. 7).

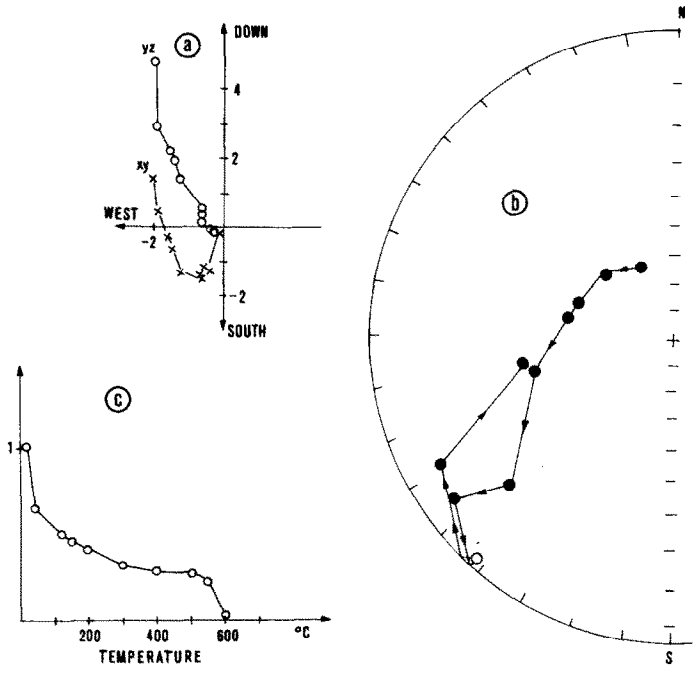


Fig. 4. NRM during thermal demagnetization of a typical specimen of mixed polarity group. For explanations see Fig. 2. (After bedding correction.)

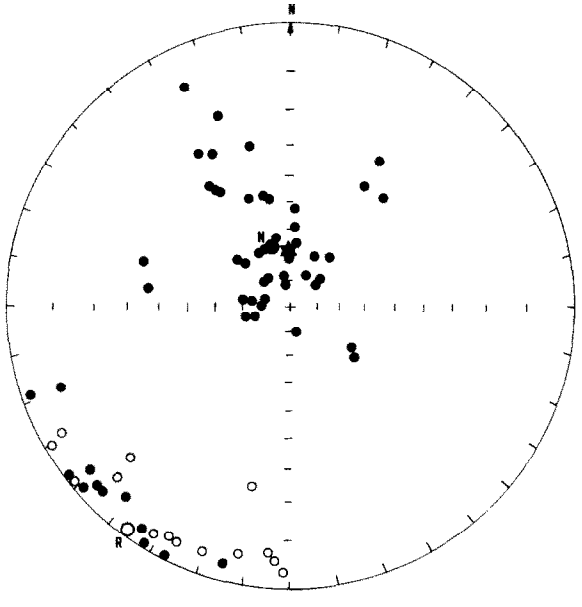


Fig. 5. Characteristic components of NRM from Bukowa Quarry. N group (before bedding correction) and R group (after bedding correction) are shown. The present-day geomagnetic field is shown by an asterisk. Open (solid) circles represent upper (lower) hemisphere directions.

To explain the observed remanence vector distribution pattern, one should examine the relationship between the declination and blocking temperature within the R group in detail. As shown in Fig. 6, the vectors which were removed below 600 °C lie more westward than those associated with the carriers of the highest (around 650 °C) blocking temperatures. Thus, the westward subgroup, called here RM, comprises magnetic carriers of distinctly different blocking temperatures from the eastward (RH) subgroup. The results of thermomagnetic analysis (Fig. 8) confirm that a higher temperature (650 °C) is required to unblock the RH subgroup remanence than the RM subgroup (610 °C) remanence. This difference in blocking temperature spectra (T_b) between the RH and RM subgroups may also be seen by comparing the demagnetization curves in Fig. 3. As the T_b range in both cases is characteristic for hematite, the presence of two generations of this magnetic carrier is postulated.

The acquisition time of the NRM components is unclear. The N group directions are of a viscous type, with a blocking temperature not exceeding

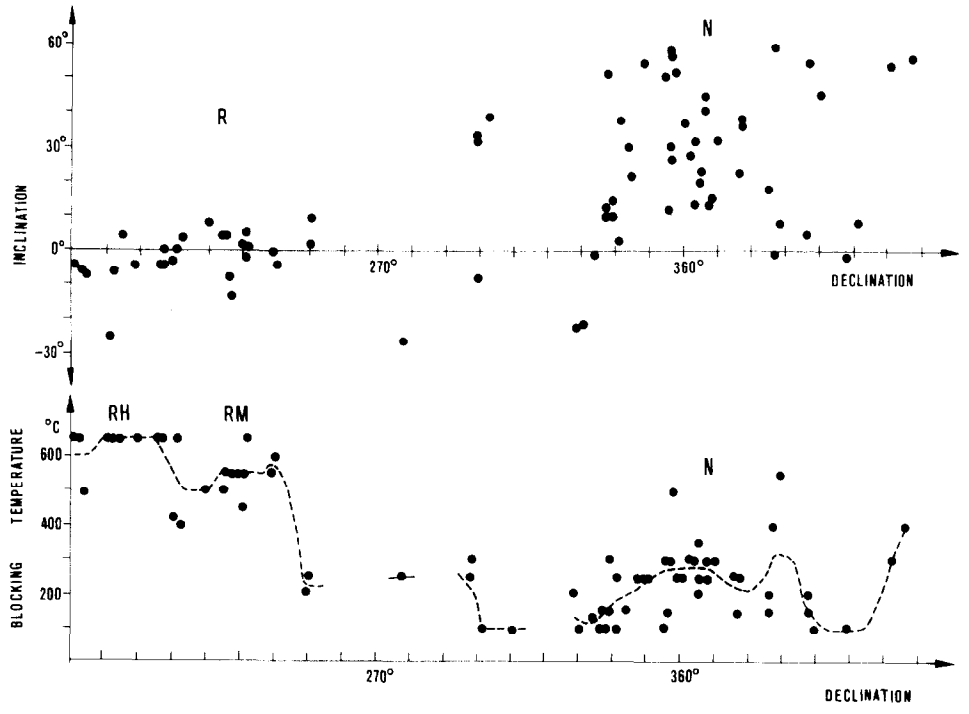


Fig. 6. Distribution of the characteristic components of NRM (upper figure) with corresponding max. blocking temperatures (lower figure). *R*—reversed magnetization, *N*—normal magnetization; RH and RM subgroups (see text) are shown. All directions after bedding correction. The broken line shows mean values of maximum blocking temperatures, calculated for 10° intervals of declination with a 5° step.

350 °C. Thermomagnetic analysis suggests the presence of magnetite in some samples (maximum T_b of the order of 580 °C—see Fig. 8c). The coincidence of N directions, before bedding correction, with the present-day geomagnetic field (see Table 1 and Fig. 5) supports the hypothesis of a secondary origin for this group. The remanence of the R group, carried by hematite, is of two-stage origin, but the time interval between the stages is difficult to estimate. The characteristic direction of the RH subgroup before bedding correction is almost identical (see Fig. 9) with the characteristic NRM obtained for nearby red beds of Permian age (Lewandowski, 1983). Therefore, the same Permian age might be presumed for the acquisition of RH magnetization. After bedding correction, the discussed direction is placed off the known paleomagnetic directions. On the contrary, tilt correction applied to the RM subgroup characteristic direction places it close to known Middle and Upper Devonian poles for Europe (Fig. 9). The hematite carrying the RM magnetization could be explained in terms of oxidation of magnetite grains soon after deposition, thus recording the ambient geomagnetic field of Middle–Upper Devonian time before the beds were tilted. It should be stressed that the pole obtained for the R population is also close to European Middle–Upper Devonian poles; however, Fisherian statistics

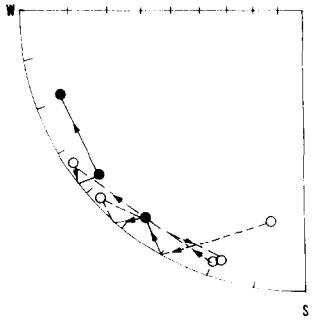


Fig. 7. Changes of sample mean directions shown from the oldest to the youngest in their chronological succession. Open (solid) circles are on the upper (lower) hemisphere of the stereonet.

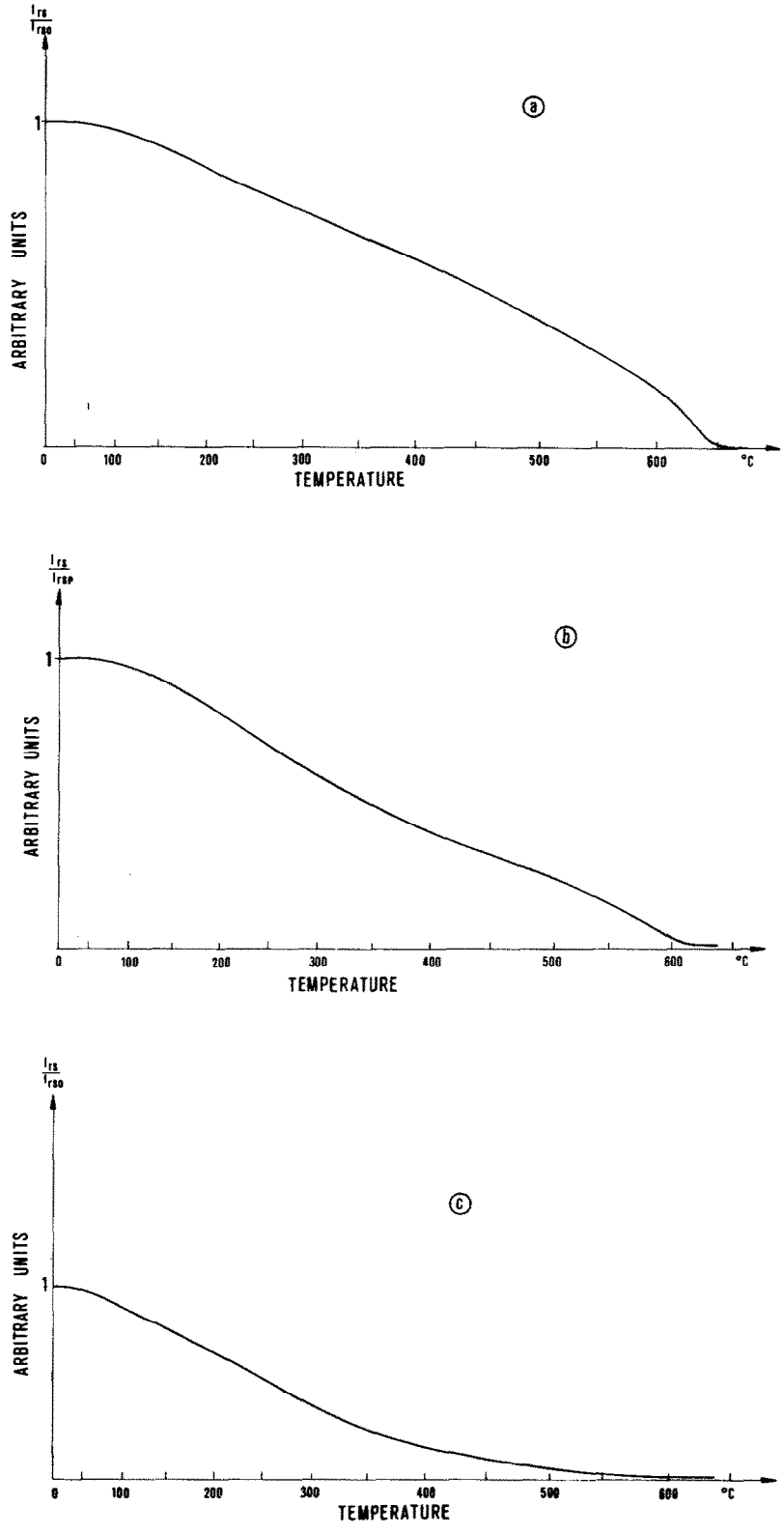


Fig. 8. Thermomagnetic curves for (a) RH subgroup, (b) RM subgroup and (c) N group representatives. I_{rso}—initial saturation remanence, I_{rs}—saturation remanence in given temperature.

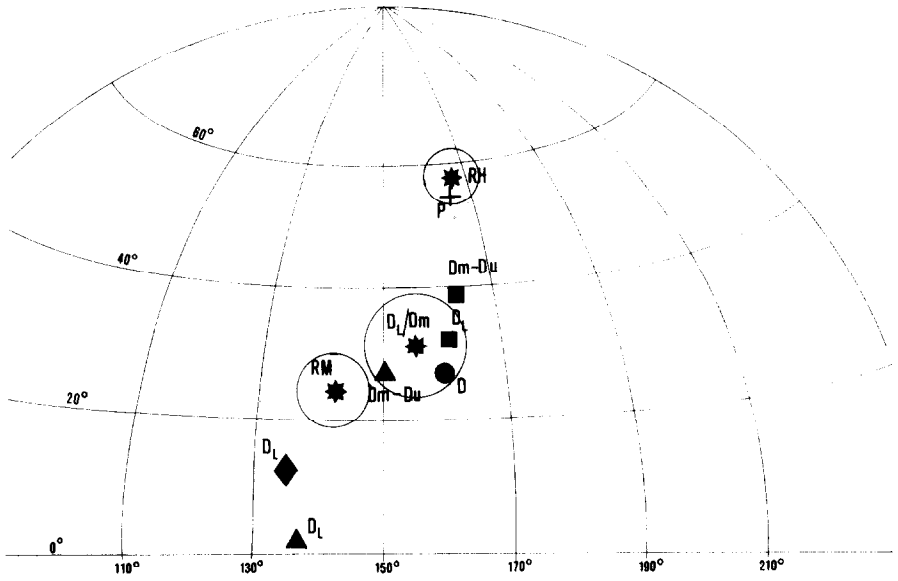


Fig. 9. Pole positons from Bukowa Quarry, calculated for R group and for RM and RH subgroups (asterisks denoted by corresponding symbols with circles of 95% confidence), compared with mean Devonian poles for Europe. Key: triangles—British Islands; squares—Russian Platform; dot—Continental Europe. (From Irving et al. (1976), Hicken et al. (1976), and McElhinny and Cowley (1977, 1980).) Diamond—Dicksonfiorden (Svalbard, after Jelenska and Lewandowski (1986), cross—Permian of Holy Cross Mts. (after Lewandowski, 1983).

TABLE 1

Characteristic mean directions and pole positions for Emsian sandstones of Bukowa Quarry, Holy Cross Mountains *

Population	n	N	Declination (degrees)	Inclination (degrees)	k	$\alpha_{95}{}^0$	Degrees N	Degrees E	Tilt correction
N	44	26	358	32	5	9.6	57	204	applied
	44	26	342	64	8	8.3	77	277	not applied
R	26	13	217	−2	14	7.8	32	156	applied
	26	13	221	−34	14	7.8	45	140	not applied
RM	15	9	231	0	39	6.2	24	142	applied
	15	9	236	−28	24	7.9	33	128	not applied
RH	11	4	197	−6	38	7.5	40	178	applied
	11	4	199	−38	51	6.5	57	166	not applied

* n—number of specimens; N—number of hand samples; sampling locality 50.5° N, 20.3° E.

applied to the directions of the R group as a whole seems to be inappropriate.

To elucidate the origin of the NRM in the rocks under discussion, an adequate fold test, together with further mineralogical investigations, are necessary, and these will be the subject of further study.

Conclusions

Three components of the ancient magnetization, recorded in Emsian quartz sandstones, have been established:

(1) a low-temperature component of secondary origin and most probably of Cenozoic age;

(2) a higher-temperature component, residing in hematite grains, thought to represent the Devonian paleofield;

(3) the highest-temperature component, of uncertain age, recorded in a second generation of hematite grains.

The pole position of the RM group from Bukowa Quarry correlates with European pole positions of Devonian age (Fig. 9). However, comparison of the RM pole with the Dicksonfjorden (Svalbard) Old Red Sandstone pole (Jeleńska and Lewandowski, in press) reveals a tangible discrepancy in position between the poles. This suggests possible post-Devonian translation of the Spitsbergen area relative to Holy Cross Mountains. The sense and magnitude of this relative displacement require further study.

Acknowledgments

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